

Rice Varietal Distribution in Sri Lanka - 2016



**Rice Research and Development Institute
Department of Agriculture**

June 2017



This volume provides the rice varietal distribution in Sri Lanka for the year 2016, prepared using information pertaining to Maha 2015/16 and Yala 2016 seasons.

Highlights ...

- ❖ The most popular rice variety is Bg 352 (16.4% of area) followed by Bg 300 (15.4% of area).
- ❖ Bg 352 and Bg 300 are widely cultivated in Mahaweli areas and Kurunegala district respectively.
- ❖ 3 ½ months maturity age group is adopted in 69.8% of the area cultivated.
- ❖ Traditional varieties are adopted in 0.45% of the area cultivated. (a reduction of 0.01% compared to 2015).
- ❖ White-pericarp varieties are adopted in 81% of area cultivated.
- ❖ Long grain rice varieties are adopted in 73% of area cultivated.

New Rice Varieties Released in 2016

Bg 252



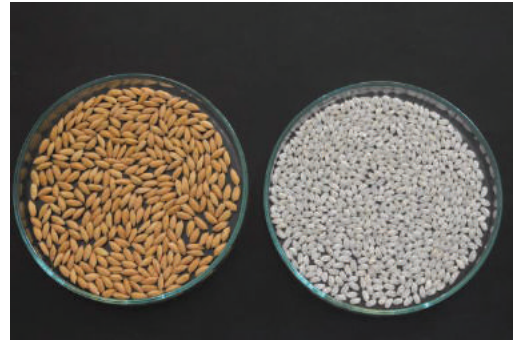
- Selection from Bg 250
- Maturity duration: 85 days
- Suitable for drought prone areas
- Average yield: 4.5 t/ha
- Potential yield: 9.95 t/ha (Murunkan)
- Short round red rice variety
- No lodging (Suitable for mechanical harvesting)

Ld 253



- At 04 selection
- Maturity duration: 85 days
- Suitable for general cultivation
- Average yield:
DZ - 5.5 t/ha
WZ - 3.5 -4.5 t/ha
- Long slender white pericarp variety
- Good taste soft cooked rice
- No seed discoloration in wet zone

Bg 374



- Pedigree:12-38-01/ Bg 360
- Maturity duration:105 days
- Suitable for general cultivation
- Average yield: 5.3 t/ha
- Potential yield: 10.5 t/ha
- Intermediate bold white pericarp variety
- High yield potential

Table 1: Summary of rice varietal distribution in Sri Lanka – 2016

Age Group	Variety	Grain type	Maha 2015/16		Yala 2016		Annual	
			(ha)	%	(ha)	%	(ha)	%
New Improved Varieties								
2 1/2 months	Bg 750	Red,LG	27	0.003	8	0.002	35	0.003
	Bg 250	White,LG	3,304	0.422	3,571	0.890	6,875	0.581
	Other		0	0.000	4	0.001	4	0.000
	Sub Total		3,331	0.426	3,583	0.893	6,914	0.584
3 months	Bw272-6b	Red,LG	931	0.119	388	0.097	1,319	0.111
	Bg 300	White,LG	109,675	14.015	72,229	18.000	181,904	15.366
	Bg 301	Red,LG	1,084	0.139	0	0.000	1,084	0.092
	At 303	Red,LG	2,807	0.359	2,317	0.577	5,124	0.433
	Bg 304	White,LG	5,021	0.642	8	0.002	5,029	0.425
	Bg 305	White,LG	5,776	0.738	411	0.102	6,187	0.523
	At 306	White,LG	1,132	0.145	152	0.038	1,284	0.108
	At 307	White,LG	22,685	2.899	8,533	2.127	31,218	2.637
	At 308	White,SG	29,799	3.808	12,080	3.011	41,879	3.538
	Bg 310	White,LG	70	0.009	34	0.008	104	0.009
	Other		115	0.015	63	0.016	178	0.015
	Sub Total		179,095	22.886	96,214	23.978	275,310	23.256
3 1/2 months	Bg 94 -1	White,LG	37,283	4.764	28,897	7.201	66,179	5.590
	Bg 94 -2	White,LG	73	0.009	0	0.000	73	0.006
	Bw 267-3	White,LG	193	0.025	14	0.004	207	0.017
	Bg 350	Red,LG	512	0.065	470	0.117	982	0.083
	Bw 351	Red,LG	261	0.033	1	0.000	262	0.022
	Bg 352	White,LG	119,678	15.293	74,614	18.595	194,293	16.412
	At 353	Red,LG	9,497	1.214	2,448	0.610	11,945	1.009
	At 354	White,LG	65	0.008	20	0.005	85	0.007
	Ld 355	White,SG	30	0.004	0	0.000	30	0.003
	Ld 356	Red,SG	1,918	0.245	53	0.013	1,971	0.166
	Bg 357	White,LG	13,361	1.707	5,703	1.421	19,065	1.610
	Bg 358	White,SG	57,876	7.396	28,895	7.201	86,772	7.330
	Bg 359	White,LG	38,906	4.972	16,036	3.996	54,941	4.641
	Bg 360	White,SG	54,368	6.947	40,599	10.118	94,967	8.022
	Bw 361	Red,LG	1,320	0.169	724	0.181	2,044	0.173
	At 362	Red,LG	99,899	12.766	57,586	14.351	157,485	13.303
	Bw 363	White,LG	89	0.011	966	0.241	1,055	0.089
	Bw 364	Red,LG	3,171	0.405	1,054	0.263	4,226	0.357
	Ld 365	Red,SG	9,481	1.212	1,670	0.416	11,152	0.942
	Bg 366	White,LG	39,696	5.073	17,859	4.451	57,555	4.862
	Bw 367	White,SG	26,157	3.342	7,063	1.760	33,220	2.806
	Ld 368	Red,SG	5,892	0.753	614	0.153	6,506	0.550
	Bg 369	White,LG	2	0.000	6	0.001	8	0.001
	Bg 370	White,SG	623	0.080	695	0.173	1,319	0.111
	Ld 371	White,SG	28	0.004	16	0.004	44	0.004
	Bw 372	Red,LG	151	0.019	264	0.066	414	0.035
	Other		15,942	2.037	2,898	0.722	18,840	1.591
	Sub Total		536,472	68.553	289,167	72.064	825,639	69.743

Age Group	Variety	Grain type	Maha 2015/16		Yala 2016		Annual	
			(ha)	%	(ha)	%	(ha)	%
4-4 1/2 months	Bg 379-2	White,LG	13,607	1.739	3,136	0.782	16,743	1.414
	Bg 400-1	White,LG	404	0.052	60	0.015	464	0.039
	At 401	Red,LG	3	0.000	0	0.000	3	0.000
	At 402	Red,LG	303	0.039	0	0.000	303	0.026
	Bg 403	White,LG	3,892	0.497	1,062	0.265	4,954	0.418
	At 405	White,LG	405	0.052	0	0.000	405	0.034
	Bg 406	Red,LG	10,501	1.342	100	0.025	10,601	0.895
	Ld 408	Red,LG	141	0.018	5	0.001	146	0.012
	Bg 450	White,SG	3,962	0.506	1,019	0.254	4,981	0.421
	Bw 451	White,LG	0	0	6	0.001	6	0.001
	Bw 452	Red,LG	0	0	10	0.002	10	0.001
	Bg 454	White,LG	17	0.002	1	0.000	18	0.002
	Bg 455	Red,LG	7	0.001	1	0.000	8	0.001
	Pokuru Samba*	White,SG	21,876	2.795	4,799	1.196	26,675	2.253
	Other		1,763	0.225	969	0.241	2,732	0.231
	Sub Total		56,880	7.268	11,169	2.783	68,049	5.748
5-6 months	Bg 745	White,SG	82	0.011	78	0.019	160	0.014
	Bg 3-5	White,LG	22	0.003	58	0.014	80	0.007
	Bg 38	White,SG	95	0.012	1	0.000	96	0.008
	Bg 407 H	White,LG	106	0.014	0	0.000	106	0.009
	Other		504	0.064	54	0.013	558	0.047
	Sub Total		810	0.103	191	0.047	1,000	0.084
Old Improved Varieties								
31/2 months	CO-10	Red,LG	36	0.005	0	0.000	36	0.003
	Sub Total		36	0.005	0	0.000	36	0.003
4-4 1/2 months	H4	Red,LG	1,220	0.156	299	0.075	1,520	0.128
	Other		14	0.002	59	0.015	73	0.006
	Sub Total		1,234	0.158	358	0.089	1,593	0.135
Traditional Rice Varieties								
3 months	Pachchaperumal	Red,LG	131	0.017	55	0.014	186	0.016
	Other		59	0.008	18	0.004	77	0.006
	Sub Total		190	0.024	73	0.018	263	0.022
3 1/2 months	Suwadel	White,SG	679	0.087	211	0.053	891	0.075
	Kalu Heenati	Red,LG	66	0.008	78	0.019	144	0.012
	Madathawalu	Red,LG	70	0.009	13	0.003	82	0.007
	Suduru Samba	White,SG	0	0	3	0.001	3	0.000
	Kahawanu	White,SG	3	0.000	0	0.000	3	0.000
	Sinna Vellai	Red,LG	9	0.001	0	0.000	9	0.001
	Batapolaal	Red,LG	7	0.001	1	0.000	8	0.001
	Ranthebili	White,SG	1	0.000	0	0.000	1	0.000
	Other		375	0.048	98	0.025	474	0.040
	Sub Total		1,210	0.155	404	0.101	1,614	0.136

Age Group	Variety	Grain type	Maha 2015/16		Yala 2016		Annual	
			(ha)	%	(ha)	%	(ha)	%
4-4 1/2 months	Kuruluthuda	Red,LG	200	0.026	61	0.015	261	0.022
	Sudu Heenati	Red,LG	55	0.007	5	0.001	60	0.005
	Moddaikaruppan	Red,LG	1,004	0.128	0	0.000	1,004	0.085
	Kahamaala	Red,LG	10	0.001	3	0.001	13	0.001
	Other		573	0.073	23	0.006	596	0.050
	Sub Total		1,841	0.235	92	0.023	1,933	0.163
5-6 months	Gonabaru	White,SG	4	0.001	3	0.001	7	0.001
	Maa wee	Red,LG	27	0.003	0	0.000	27	0.002
	Rathel	White,SG	44	0.006	7	0.002	51	0.004
	Murunkan	Red,LG	643	0.082	0	0.000	643	0.054
	Periya Velladiyan	Red,LG	291	0.037	0	0.000	291	0.025
	Other		452	0.058	2	0.001	455	0.038
	Sub Total		1,462	0.187	12	0.003	1,474	0.125
Total			782,561	100.000	401,263	100.000	1,183,824	100.000

Note: LG=Long Grain (Nadu), SG=Short Grain (Samba), *Not a recommended rice variety

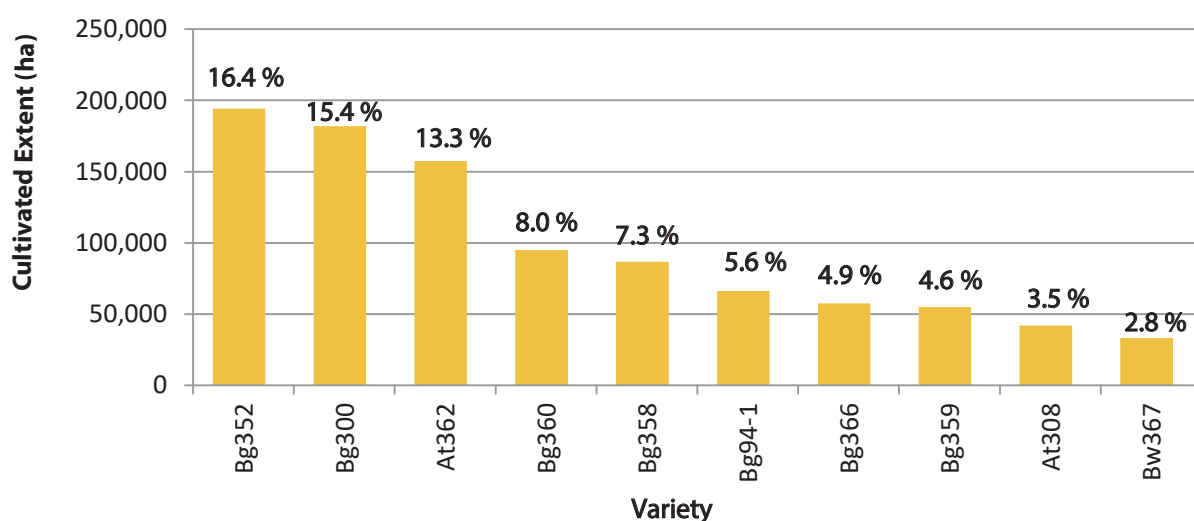


Figure 1 : Cultivated extents of ten most popular rice varieties in Sri Lanka - 2016

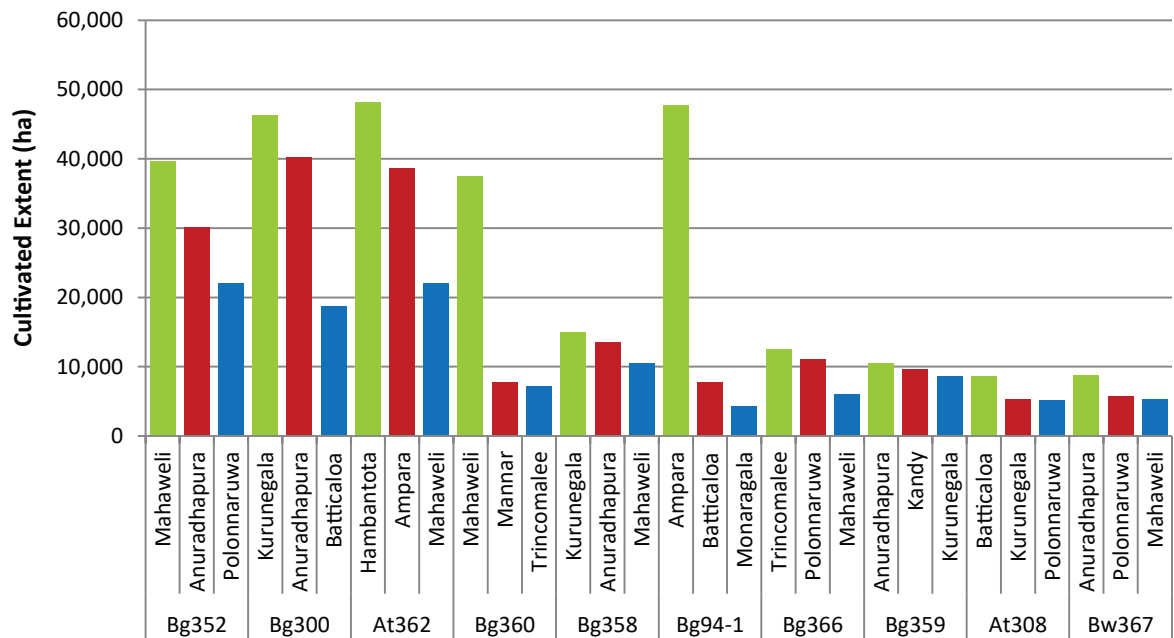


Figure 2 : Cultivated extents of ten most popular rice varieties in major three districts in Sri Lanka 2016

Table 2 : Cultivated extent of rice by variety category - 2016

Category	Maha 2015 /16		Yala 2016		Annual Extent	
	(ha)	%	(ha)	%	(ha)	%
New Improved Varieties	776,589	99.24	400,323	99.77	1,176,912	99.42
Old Improved Varieties	1,270	0.16	358	0.09	1,629	0.14
Traditional Rice Varieties	4,703	0.60	581	0.14	5,284	0.45

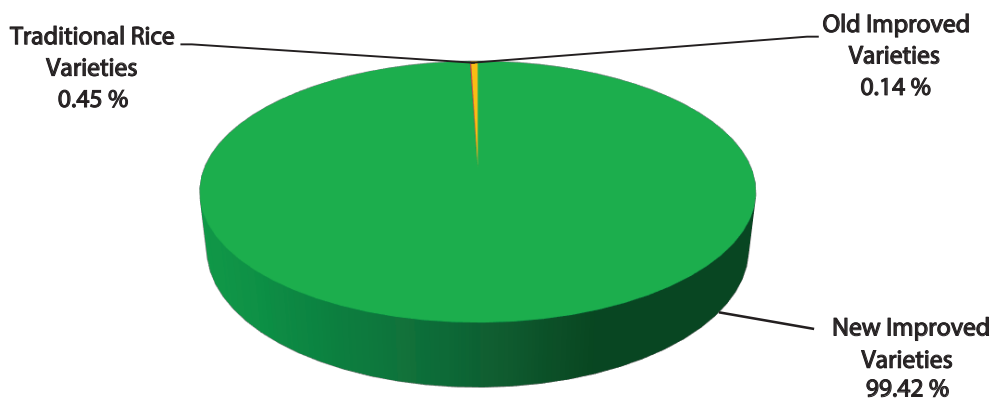
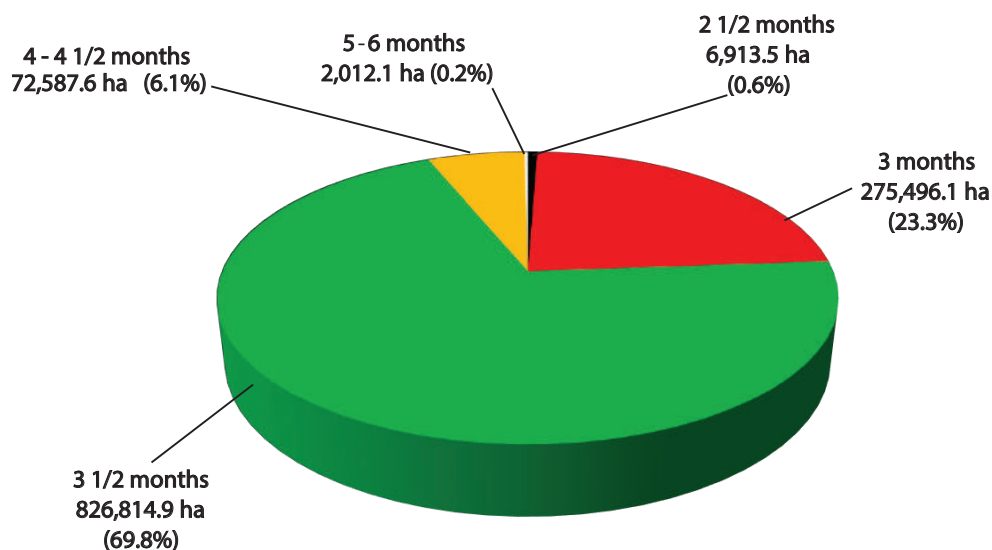
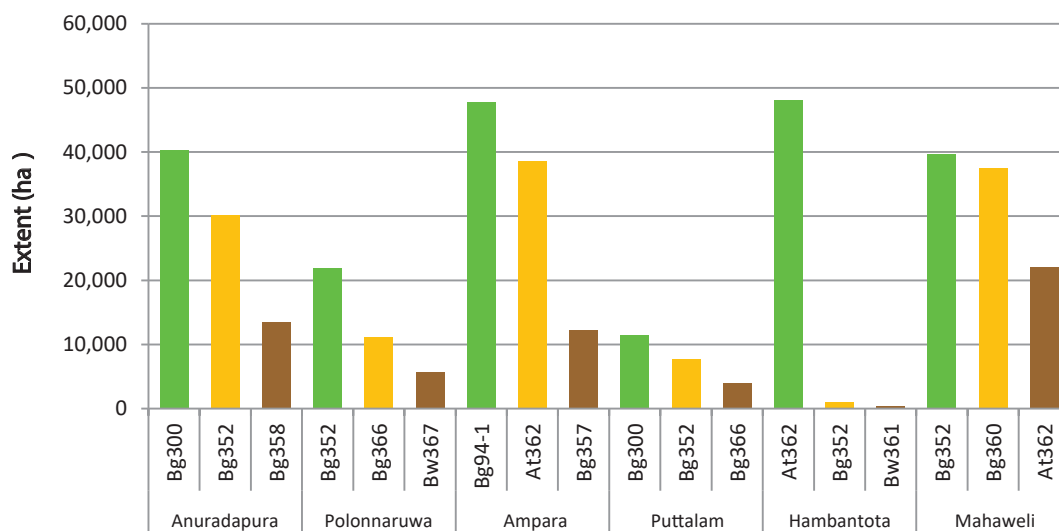


Figure 3 : Cultivated extent of rice by variety category - 2016

Table 3 : Varietal distribution of rice varieties according to age group - 2016

Age Group	Maha 2015/16		Yala 2016		Annual Extent	
	(ha)	%	(ha)	%	(ha)	%
2 ½ Months	3,331	0.43	3,583	0.89	6,914	0.58
3 Months	179,286	22.91	96,287	24.00	275,573	23.28
3½ Months	537,718	68.71	289,571	72.16	827,289	69.88
4-4½ Months	59,960	7.66	11,622	2.90	71,582	6.05
5-6 Months	2,267	0.29	200	0.05	2,467	0.21

**Figure 4 : Variety distribution of new improved varieties according to age group - 2015****Figure 5 a : Spread of three major rice varieties in each district including Mahaweli areas in Dry Zone- 2016**

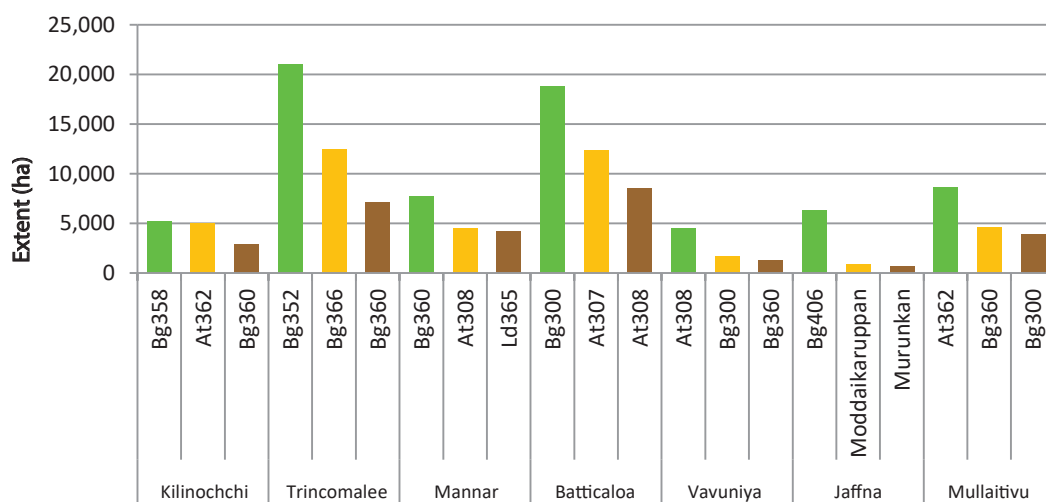


Figure 5 b : Spread of three major rice varieties in each district including Mahaweli areas in Dry Zone- 2016

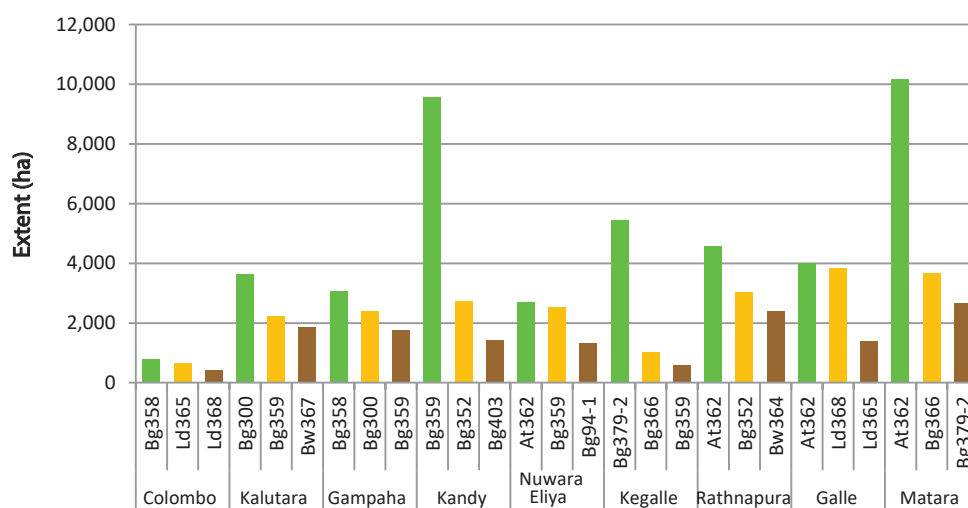


Figure 6 : Spread of three major rice varieties in Wet Zone - 2016

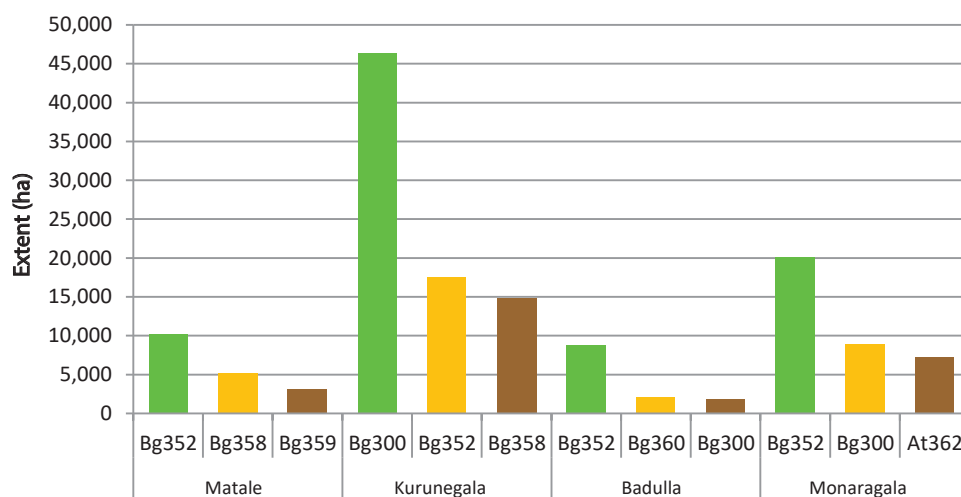


Figure 7: Spread of three major rice varieties in Intermediate Zone - 2016

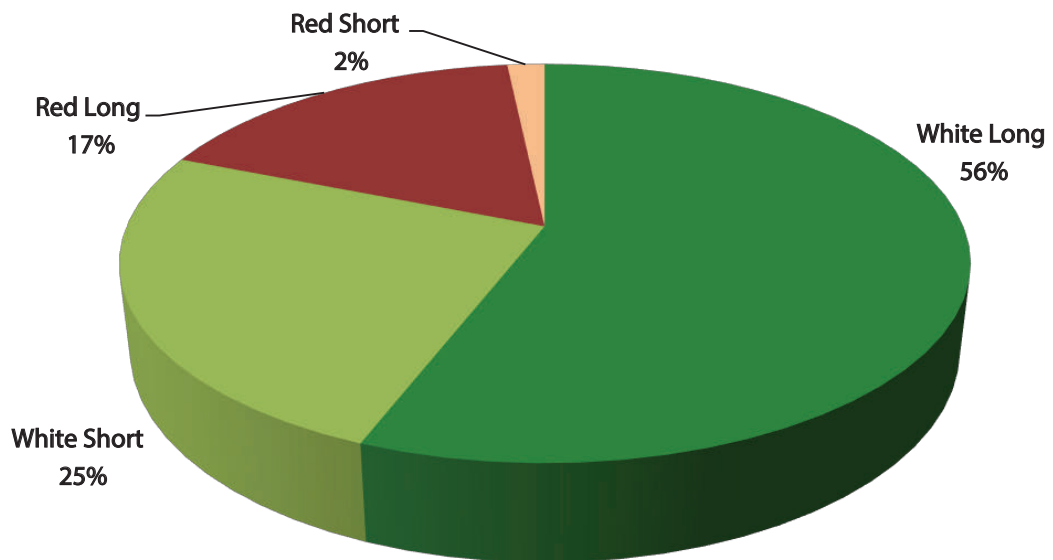
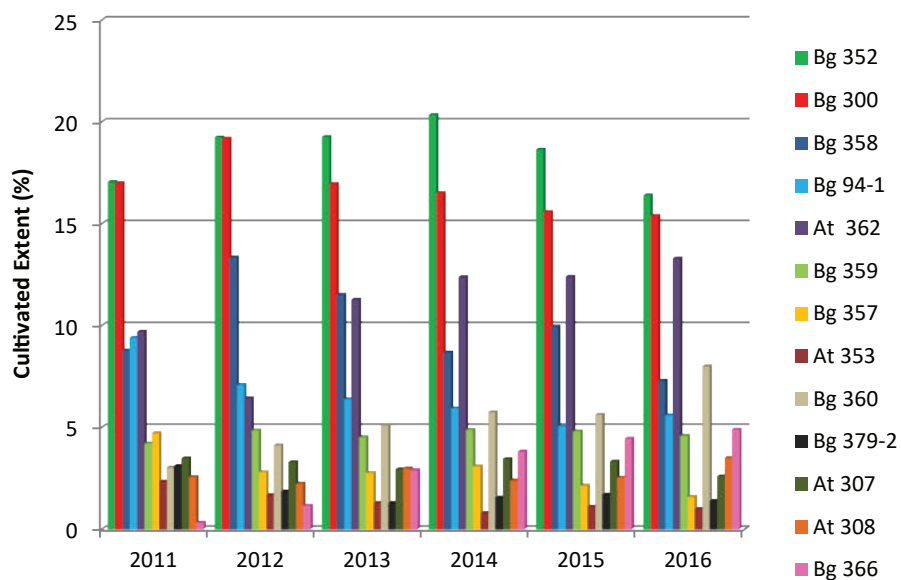


Figure 8 : Varietal spread according to grain type - 2016



Note: Popular rice varieties in last five years relative to 2011

Figure 9 : Spread of popular rice varieties in Sri Lanka (2011 - 2016)

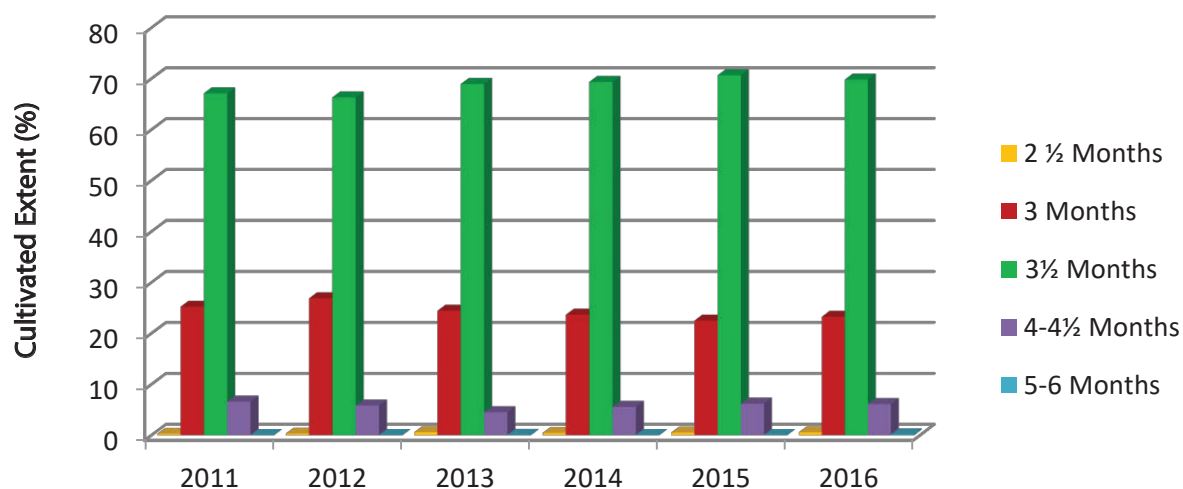


Figure 10 : Distribution of rice varieties by different age group 2011 - 2016

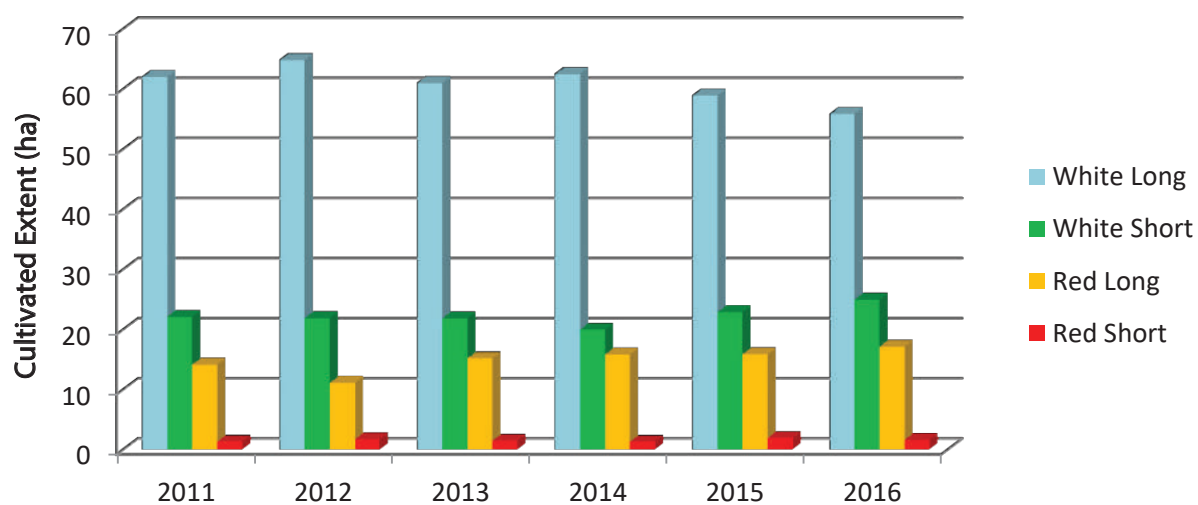


Figure 11: Distribution of rice varieties by grain type and colour 2011 - 2016

Table 4 : New rice varieties released by RRDI after 2005

Year of release	Variety	Maturity group	Grain size and colour	Area of recommendation
2005	Bg 250	2½	White-IB	Drought and flood prone areas (as escape)
2005	At 307	3	White-IB	General cultivation
2005	Bg 406	4 - 4½	Red -IB	Northern region
2005	Bg 407H	4 - 4½	White-IB	High potential areas (1 st hybrid variety)
2005	Bg 454	4 - 4½	White-IB	General cultivation with assured supply of water
2006	Bw 364	3½	White-IB	Wet zone- Tolerant to iron toxicity
2008	Ld 365	3½	Red-SR	Wet zone
2008	At 308	3	White-SR	General cultivation
2009	Bg 366	3½	White - IB	General cultivation
2010	Ld 408	4 - 4½	Red - IB	General cultivation
2011	Bw 367	3 ½	White-SR	General cultivation -Tolerant to iron toxicity
2011	Ld 368	3½	Red-SR	Wet zone
2012	Bg 369	3½	White-LM	Salinity prone areas
2013	Bg 370	3½	White-SR	General cultivation
2013	Ld 371	3½	White-SR	Wet zone
2013	Bw 372	3½	Red - IB	Wet zone-Tolerant to iron toxicity
2013	At 309	3	White-LS	General cultivation - For rice based products
2014	At 373	3½	White-SR	General cultivation - Ambalantota Suwanda Samba
2014	Bg 251	2½	White-IB	Drought prone areas
2014	Bg 310	3	White-IB	General cultivation-Salinity tolerant
2014	Bg 455	4½	Red -IB	Flood prone areas
2015	At 311	3	Red-LS	General cultivation - Neeroga
2016	Bg 252	2½	Red-SR	Drought prone areas
2016	Ld 253	2½	White-LS	General cultivation
2016	Bg 374	3½	White-IB	General cultivation

Long grain (Nadu) : LM (Long Medium), LS (Long Slender), IB (Intermediate Bold)
Short grain (Samba) : Short Round

Notes

Message from Director RRDI....

Rice Research and Development Institute is the national center for the development and primary dissemination of rice technologies to improve the productivity and profitability of rice farming and quality of rice. In order to achieve the stated objective, a need of a sound database on rice varietal distribution is of vital importance.

RRDI has been on the process of estimating the rice varietal distribution of Sri Lanka for many decades with the cooperation of Provincial and Inter-provincial Department of Agriculture and Mahaweli Authority of Sri Lanka.

This newsletter on varietal distribution is highly used by scientists, students, academia and policy planners. I greatly appreciate the effort made by Socio Economics Division of Rice Research and Development Institute for this endeavour.

Dr. Amitha P. Bentota

Director (Rice Research and Development)

Prepared by

J.K. Galabada, Agricultural Economist, RRDI

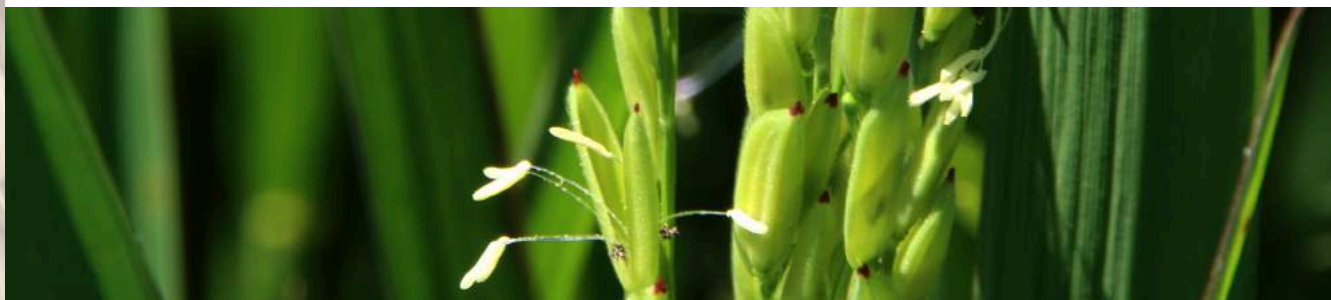
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With courtesy of

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